Chun-Wei Chiang

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Summary

Applied Scientist with a PhD in Computer Science specializing in LLM evaluation methodology, post-training, agentic systems, and human behavior modeling. Shipped production systems that simulate human populations with LLMs and rigorously measure their fidelity.

Professional Experience

Qualtrics June 2025 - Present

Applied Scientist II

- Led modeling & evaluation for Synthetic Sampling from pre–Early Access to General Availability, post-training LLMs on **7M+ human survey responses** (LoRA, FSDP) to simulate survey populations; **named inventor on 2 patents pending**.
- Built the framework measuring the model's fidelity to real respondents: across 28 test surveys, it aligns with human response distributions on **nearly 4x as many questions as a frontier LLM**.
- Owned its science evaluation: the agent covers **84% of the research questions** answered in 10+ human-written reports; built hallucination checks tracing each reported number back to a tool call.

Purdue University January 2020 - May 2025

Graduate Research Assistant

- Developed evaluation frameworks combining controlled human-subject experiments with behavioral metrics to measure human-AI decision-making outcomes across 12+ published studies.
- Built an LLM-based agent architecture for real-time conversational intervention in multi-party discussions, determining target recipient, timing, and response strategy via dialogue-state modeling.

Amazon (Twitch) June 2023 - August 2023

Applied Science Intern

- Developed and evaluated a transformer-based collaborative filtering and ranking model, improving recommendation precision by 15% in pilot experiments.

Honda Research Institute January 2023 - May 2023

Research Scientist Intern

- Researched driver behavior and user experience in autonomous vehicles and conventional vehicles, leveraging quantitative analysis to design safer, more user-friendly interfaces.

Brain Technologies June 2022 - August 2022

Natural Language Processing (NLP) Research Intern

- Applied LLM and collaborative filtering techniques to improve recommendation relevance, including web RAG-based retrieval for dynamic knowledge integration.

Education

Ph.D. in Computer Science, Purdue University January 2020 - May 2025

Skills

Machine Learning / LLMs	LLM evaluation & calibration, LLM-as-a-judge, post-training (LoRA/SFT), human preference & behavior modeling, distributed training (FSDP), agentic systems (LangGraph Deep Agents, MCP)
Programming & Data	Python, SQL, Java, R, PostgreSQL, MySQL, MongoDB, Pandas
Frameworks & Tools	PyTorch, TensorFlow, Hugging Face, scikit-learn, Git, AWS

Honors

Outstanding Reviewer Recognition (CHI, CSCW, DIS); Best Poster Honorable Mention (WWW).

Selected Publications

12+ peer-reviewed publications in top-tier venues (CHI, WWW, IUI, CSCW, IJCAI); 900+ citations, h-index 12. Google Scholar

- **Chun-Wei Chiang**, et al. Enhancing AI-Assisted Group Decision Making through LLM-Powered Devil's Advocate. *IUI 2024*. (200+ citations)
- **Chun-Wei Chiang**, et al. Are Two Heads Better Than One in AI-Assisted Decision Making? *CHI 2023*. (100+ citations)